

Cox-Ingersoll-Ross (CIR) Process Age Inference Algorithm for Parameter Estimation Manual (v.1.0)

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1. Introduction

The MATLAB software for the Cox-Ingersoll-Ross (CIR) Process Age Inference Algorithm is designed to estimate CIR process model parameters (α , β , and ρ) from a set of radiocarbon records. We regard this software as the first half of the next version of BIGMACS [Lee and Rand et al. (2023), “Bayesian age models and stacks: combining age inferences from radiocarbon and benthic $\delta^{18}\text{O}$ stratigraphic alignment”, *Climate of the Past*]; the second half will be about the stack construction, which will be available soon.

Here we provide instructions to download the software, prepare the required input files, and estimate CIR process model parameters. This software is designed to reproduce the results in [Lee et al., in review].

When you are using this software, please cite the following paper:

Lee et al. (2026), “Scalable and consistent sedimentation rate modeling via the Cox-Ingersoll-Ross (CIR) process for ocean sediment core age model development”, in review and submitted to *Geoscientific Model Development*.

Happy chronology building!

2. Installation

The latest version of the open-sourced software package can be installed from GitHub using the following link:

https://github.com/eilion/CIR_Process_Aligner_MATLAB/Trainer_MATLAB

The software was developed and tested on MATLAB R2024a and requires the statistics and machine learning toolbox and the parallel computing toolbox. An example run is included with the installation. To confirm that the software is running properly, open and run the function `main('DATA_02282025',2.0,2.0,0.5,'No',0.5,10000,'No')`. This will begin parameter estimation for the 79 radiocarbon records used in the paper [Lee et al, 2026].

3. Preparing input files

To construct age-depth models, create a new folder as `/Inputs/[inputFolder]` with the desired run name. All proxy data files should be of the txt format. See the input folder `'/Inputs/DATA_02282025'` as an example for structure and formatting. Under `/Inputs/[inputFolder]/records/`, put a set of the txt files with the record names. Each record stores the radiocarbon and calibration data for the sediment core and has the following six columns: (1) depth (in meters); (2) age (in kiloyears); (3) error (in kiloyears); (4) dR (in kiloyears); (5) dSTD (in kiloyears); and (6) cc ('1' for IntCal20, '2' for Marine20, and '3' for SHCal20), but cc is not used here.

Under `/Inputs/[inputFolder]/`, the following two txt files should be specified:

3.1. average_sedrates.txt specifies the initial standardization parameters of ^{14}C records, in meter/kiloyear.

3.2. calibration.txt specifies the calibration model, e.g., IntCal20, Marine20, SHCal20, to calibrate ^{14}C records in `/Inputs/[inputFolder]/records/`.

To configure the setting shared by all the records, you need to first copy and paste `/Defaults/setting.txt` to `/Inputs/[inputFolder]/setting.txt` and then modify the values in the txt file as you want:

- **nSamples** is the number of age-depth model samples to draw for the posterior approximation.
- **a** and **b** are the parameter values of the generalized Student's t-distributions for ^{14}C .

4. Running the algorithm

To run the algorithm to estimate CIR process model parameters, just execute the command `main([inputFolder],ALPHA,BETA0,RHO0,isINIT,INT,max_iters,itors_for_R)`, where `[inputFolder]` is the name of the folder in `/Inputs`, by following the instruction in the MATLAB function `main.m`. To draw posterior samples, just execute the command `draw_Samples([inputFolder],ALPHA,BETA0,RHO0,INT)` by following the instruction in the MATLAB function `draw_Samples.m`.

5. Output files

The outputs of running the function `/main([inputFolder],ALPHA,BETA0,RHO0,isINIT,INT,max_iters,itors_for_R)` will be stored in `/Initializations/([inputFolder]_K_[2*ALPHA]_INT_[INT]).mat`. You can track the updated BETA and RHO in the struct `setting.BETA` and `setting.RHO`, respectively. The outputs of running the function `/draw_Samples([inputFolder],ALPHA,BETA0,RHO0,INT)` will be stored in `/Outputs/[inputFolder]/KK_[2*ALPHA]_INT_[INT]/Trial_XXX/results.mat`. The struct `SAMPLES` contains the inverse sedimentation rate samples and age-depth model samples like that of `Aligner_MATLAB`.